Vishwakarma Institute of Technology

Computer Department

Assignment 1

Design and develop a responsive website with the help of any code generation AI tool like ChatGPT to calculate Electricity bill using PHP. Condition for first 50 units – Rs. 3.50/unit, for next 100 units– Rs. 4.00/unit, for next 100 units – Rs. 5.20/unit and for units above 250 – Rs. 6.50/unit. You can make the use of bootstrap as well as jQuery.

Introduction to following topics:

1. HTML Forms
2. CSS/Bootstrap
3. PHP Basics
4. Variables
5. Conditional Statements (if-else)
6. Arithmetic Operations
7. Form Handling using POST
8. Responsive Web Design

Objective

To design a responsive website that calculates the electricity bill according to the given tariff rates using PHP.

Software Required

Software	*Purpose*
XAMPP	PHP and Apache Server
Visual Studio Code	Code Editor
Google Chrome	Browser
Bootstrap 5	Responsive Design
jQuery	JavaScript Library

Installing Required Software

Step 1 Install XAMPP

a) Download- https://www.apachefriends.org/
b) Install it.
c) Open XAMPP Control Panel.
d) Start
    1. Apache
    2. MySQL (Not required in this assignment but starts if needed later.)
e) Open browser
    1. http://localhost

If the XAMPP page opens successfully, PHP is installed correctly.

Step 2 Install VS Code

a) Download-https://code.visualstudio.com

Install the following extensions

- ❖ PHP Intelephense
- ❖ Live Server
- ❖ Bootstrap 5 Snippets
- ❖ PHP Extension Pack

Folder Structure

1. Inside
   C:\xampp\htdocs
2. Create
   ElectricityBill
3. Inside this folder
   ElectricityBill
   index.php
   style.css

Problem Statement

Calculate electricity bill according to the following tariff.

Understanding the Logic

This is the most important part.

Units	Rate
First 50 Units	**₹3.50/unit**

Next 100 Units	**₹4.00/unit**
Next 100 Units	**₹5.20/unit**
Above 250 Units	**₹6.50/unit**

A. Suppose
Units = 40
Bill 40 × 3.5=140

B. Suppose
Units = 80
First 50 units
50 × 3.5
Remaining
30 × 4
Total
175+120=295

C. Suppose
Units=170
First 50
50 ×3.5 =175
Next 100
100×4=400
Remaining
20×5.2=104
Total
679

D. Suppose
Units=300
First 50
175
Next 100
400

Next 100

520

Remaining

50×6.5=325

Total

1420

Algorithm

1. Start
2. Accept units
3. If units **≤50**

Calculate bill

4. Else if units **≤150**

Calculate

50×3.5

+

Remaining×**4**

5. Else if units **≤250**

Calculate

50×3.5

+

100×4

+

Remaining×**5.2**

6. Else

Calculate

50×3.5

+

100×4

+

100×5.2

+

Remaining×**6.5**

7. Display Bill
8. Stop

Responsive Website Design

We can use Bootstrap Card.

A Bootstrap Card is a pre-designed container (box) provided by Bootstrap that is used to display related information in a neat, attractive, and responsive format.

Reference Code

```
<!DOCTYPE html>
<html>
<head>
   <title>Electricity Bill Calculator</title>
   <!-- Bootstrap CSS -->
   <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css
" rel="stylesheet">
</head>

<body class="bg-light">
<div class="container mt-5">
<div class="row justify-content-center">
<div class="col-md-6">
<div class="card shadow">
<div class="card-header bg-primary text-white text-center">
<h3>Electricity Bill Calculator</h3>
</div>
<div class="card-body">
<form method="POST">

<label class="form-label">
Enter Electricity Units
</label>

<input
type="number"
name="units"
class="form-control"
required>
<br>

<input
```

```
type="submit"
class="btn btn-success w-100"
value="Calculate Bill">
</form>

<?php
if(isset($_POST['units']))
{
$units=$_POST['units'];
$bill=0;
if($units<=50)
{
$bill=$units*3.5;
}
elseif($units<=150)
{
$bill=(50*3.5)+(($units-50)*4);
}
elseif($units<=250)
{
$bill=(50*3.5)+(100*4)+(($units-150)*5.2);
}
else
{
$bill=(50*3.5)+(100*4)+(100*5.2)+(($units-250)*6.5);
}
echo "<div class='alert alert-success mt-4'>";
echo "<h4>Total Bill = ₹ ".$bill."</h4>";
echo "</div>";
}
?>
</div>
</div>
</div>
</div>
</div>
</body>
</html>
```

Form

<form method="POST">

When the button is clicked, data is sent to the same page using the POST method.

Input Box

<input type="number" name="units">

Accepts electricity units from the user.

Reading Data

$units=$_POST['units'];

Stores the entered units into the variable $units.

Variable

$bill=0;

Initializes the bill amount.

First Condition

if($units<=50)

If units are 50 or less.

Example

30 units

30×3.5

Second Condition

elseif($units<=150)

Example

80 Units

50×3.5+30×4

Third Condition

elseif($units<=250)

Example

170

50×3.5+100×4+20×5.2

Fourth Condition

Else Units above 250.

Display Result

echo "<h4>Total Bill = ₹".$bill."</h4>";

Displays the calculated bill.

Sample Output

Input Units 40

Output: Total Bill ₹140

Input Units 120

Output ₹455

Input Units 180

Output ₹731

Input Units 300

Output ₹1420